



Integrated Himalayan EWS – System Architecture

Binding monsoon flooding & landslides / ice–rock avalanche / GLOF / major earthquake onto the two systems Nepal already runs (working draft)

Drafted 2026-08-30 · Revised 2026-09-03 · Based on analysis of the 26 August 2026 Bhote Koshi–Trishuli flood · Not an official position

Do not build from zero. Connect the two systems that exist; add only the missing detection paths.

1 ☀ The two systems Nepal already has

Rain	DHM rainfall & river “live watch”: 280 stations
	Web telemetry since 2010. The backbone of the rainfall track
Rain	Threshold-based flood & landslide warning (since 2018)
	272 hazard areas delineated; SMS is sent when upstream rainfall crosses threshold
Rain	CBFEWS in 8 river systems · 3-day rainfall forecast
	Koshi, Karnali, West Rapti and others. Narayani is included — already present in this corridor
Rain	1 of 3 planned weather radars
	Installed in the west in 2019. Central and east: not yet
Seis	~40 broadband seismometers · ~36 accelerometers
	DMG / NEMRC, plus 8 Univ. of Tokyo–JICA stations and 10 China CEA stations

For rainfall-driven hazards, the chain from detection to dissemination is already complete. What failed here was not the absence of a system but the fact that **the system presupposes rain**.

2 🇳🇵 So why did nothing fire?

Gap 1	The rainfall track presupposes rainfall
	No rain was involved in this event (confirmed by DHM). Neither the 280 stations nor the 272 area thresholds could respond in principle
Gap 2	The seismic track presupposes earthquakes
	Associators are built on sharp arrivals. A slow-onset mass movement is discarded at association. nst = 0 in this event’s catalogue entries is the evidence
Gap 3	The two tracks are not wired together
	Even with an M5.2 on the record (GFZ: Mw 5.7), nothing carries it into the flood alerting path. Both 08:37 and 11:45 reached the catalogue only after an analyst processed them — posted a day and a half later
Gap 4	Mandates are split
	Earthquakes = DMG / flood & landslide warning = DHM / response = NDRRMA

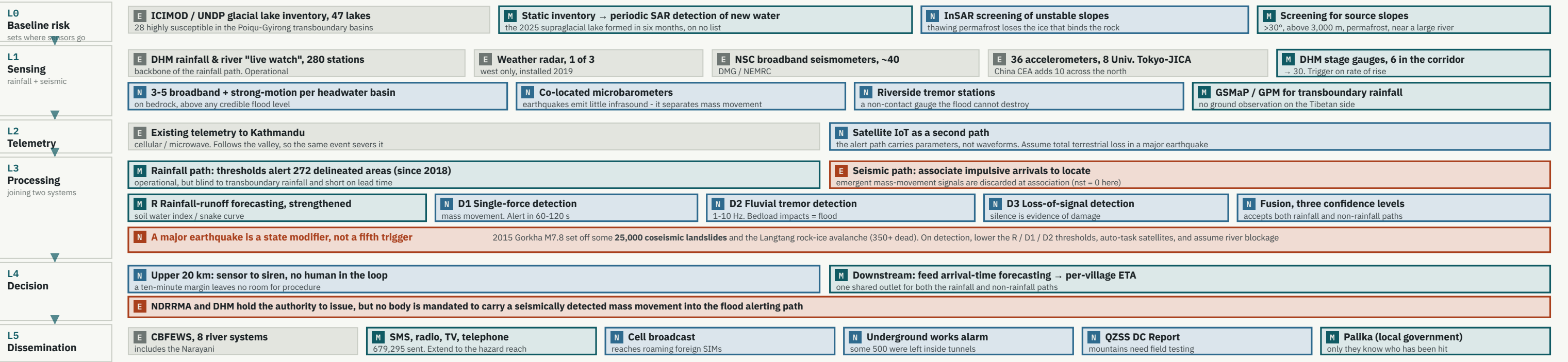
Each track works for its own hazard. Neither owned this hazard, and they were not connected to each other. That is what “integration” has to fix — and none of Gaps 1–4 is a shortage of hardware.

3 ⚡ Hazard stocktake — what is caught, and by which path

HAZARD	DETECTION PATH	STATUS
Monsoon flooding & landslides Cumulatively the largest killer of all	R rainfall threshold + soil water index	existing operating
Ice–rock avalanche / rock collapse This event = Langtang Lirung N face, bedrock and all / 2015 Langtang	D1 single-force source alert in 60–120 s	new
GLOF / landslide-dam breach This event’s initial mechanism + the barrier lake of 28 Aug	D2 fluvial tremor (1–10 Hz)	new
Loss of the instruments themselves Upstream gauges went silent together after 08:40; the 30 Aug rise was noticed by ear	D3 missing data = alarm	new
Widespread coseismic slope failure 25,000 landslides in Gorkha 2015	State modifier lowers every trigger threshold	new

What this system cannot catch — stated explicitly
Snow avalanche Detectable as a small-magnitude D1 class. Forecasting is impossible — no snowpack observation exists
Earthquake early warning itself Nepal has no national EEW (demonstration stage only)
Air blast Occurred at Langtang 2015 and Chamoli 2021. Damage reaches beyond the flow path

4 🔄 Six-layer architecture — what is reused, what is modified, what is added



Colour encodes status. Grey = existing, used as is / red outline = existing, but did not function here / teal = existing, to be modified / blue = newly added. The only permanent new installations are the three L1 items and the satellite IoT in L2. **Integration lives in L3** — the four triggers stay separate, and only the L4 arrival-time prediction and the L5 dissemination paths are shared. No single detector is bet on; each track can raise an alert independently. **Confidence has three steps** (one detector = advisory / two detectors agreeing = prepare to evacuate / downstream measurement = evacuate). **L0 is the peacetime layer**, deciding where stations go and what the thresholds are — a static glacial-lake inventory is not enough: the 2025 supraglacial lake appeared within six months.

5 🕒 Phasing — Phase 0 is the go / no-go

Phase 0	Retrospective analysis of existing archives	0–6 months · < USD 0.35 M · no hardware
	Go back through ~40 stations’ recorded waveforms and find the collapses that already happened. Threshold values, and how often a false alarm would fire, come straight out of the data.	
Phase 1	Trial running (no warnings issued)	6–18 months · < USD 0.7 M
	Run the detection alongside the existing chain. Issue no warnings, and measure only how often it would have fired in error.	
Phase 2	Pilot deployment	18–30 months · USD 1.4–2.8 M
	Fill station gaps in the Bhote Koshi–Trishuli; add infrasound and riverbank tremor sites. Automatic sirens within 20 km upstream; arrival-time prediction downstream.	
Phase 3	Roll out to other basins · transboundary integration	USD 1.4–2.1 M per basin

If Phase 0 shows events cannot be caught with adequate margin, nothing further should be funded. It costs almost nothing and governs a multi-million-dollar decision. Of 60 large Himalayan rock–ice avalanches, **about half cascaded, killing ~10× as many** (Zhong et al. 2025). The cascade is what this is for.

6 🌐 Policy challenges

1	Assign the mandate
	Widen DHM’s flood-warning remit to accept non-hydrological triggers, or stand up a joint DMG–DHM–NDRRMA cell. Without it, detection has no route to a warning.
2	Agree the false-alarm budget before go-live
	Settle a line with communities — “three false alarms a year are acceptable if no river-threatening event is missed” — and rehearse it. A system whose first false alarm comes as a surprise is switched off within a year. A political decision, not a technical one
3	Transboundary
	This source was inside Nepal , outside China’s warning system: Nepal must detect it itself. Seismic waves also cross borders, so a Tibetan-side source is caught by stations in Nepal without waiting for a data-sharing agreement — the political strength of this design
4	No footing to build on — the view from Nepal
	Hazard and risk mapping has never been done nationwide. Land-use law exists but is not enforced. Houses raised on fill from the Trishuli bed were swept away; those higher up survived. Bridges sit too low. The L0 screening is also the work of filling

What Japan can bring

NIED: Hi-net / V-net; operational automatic lahar detection

Univ. of Tokyo & JICA: 8 stations already in Nepal — a base to extend

MLIT Hokkaido Bureau: the Tokachidake sensor-to-siren direct-wiring model

PWRI / ICHARM: turning detection into arrival times and evacuation lead time

JMA & JAXA: landslide warning information (snake curve); GSMaP upstream rainfall

JAXA · Sentinel Asia · ADRC: satellite monitoring of the upstream across the border

7 🚫 Outside the scope

🏔	It does not predict. It begins once the mountain moves
	The 2.5-hour precursor at Chamoli in 2021, and the ground-temperature anomaly reported two days before this event, were both found afterwards. Important as research, but not a design assumption .
🌊	The size of the shaking is not the discharge
	Shaking is a guide to how much sediment is moving, and it has to be tuned per site. A small GLOF can hide inside a monsoon rise; what tells them apart is how fast it grows and the shape of the frequencies .
🇳🇵	Small events go unseen
	We do not yet know the smallest volume that can block a channel. Where to set the threshold is a judgement about how much risk to accept — Phase 0 turns it into a number.
📏	It is not a substitute for stage gauges
	Gauges give measured values; seismometers give survivability. They complement each other . Leaning on either alone repeats the same failure in another form.

Legend (colour coding of the architecture)

■ E = existing, used as is ■ E = existing, did not function here ■ M = existing, to be modified ■ N = newly added

New permanent hardware is limited to **the three L1 items and the L2 satellite IoT**. The rest is **software and institutions**.

Integration lives in L3 — triggers stay separate; only decision and dissemination are bound together.

Alert automatically when an event occurs.

The system does not need to decide “rock or ice”. The only question is whether a large collapse has occurred upstream. Issue the warning without waiting for the detailed analysis — pinning down the cause takes days, and here too the reading moved.